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Quantitative evaluation of anti-biofilm cavitation activity seeded from microbubbles or protein cavitation nuclei by passive acoustic mapping

Sara B Keller¹ , Gareth LuTheryn² , Michael D Gray¹ , Brian Lyons¹ , Robin O Cleveland¹ , Eleanor Stride^{1,2} and Constantin C Coussios^{1,*}

¹ Institute of Biomedical Engineering, University of Oxford, Oxford, United Kingdom

² Nuffield Department of Orthopaedics, Rheumatology and Musculoskeletal Sciences, University of Oxford, Oxford, United Kingdom

* Author to whom any correspondence should be addressed.

E-mail: constantin.coussios@eng.ox.ac.uk

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Supplementary material for this article is available [online](#)

Abstract

Objective. Bacterial biofilms represent a major challenge for effective antibiotic therapy as they confer physical and functional changes that protect bacteria from their surrounding environment. In this work, focused ultrasound in combination with cavitation nuclei was used to disrupt biofilms of *S. aureus* and *P. aeruginosa*, both of which are on the World Health Organization's priority list for new antimicrobial research. **Approach.** Single species biofilms were exposed to ultrasound (0.5 MHz center frequency, 0.5–1.5 MPa peak rarefactional pressure, 200 cycle pulses, 5 Hz repetition frequency, 30 s duration), in the presence of two different types of cavitation nuclei. Quantitative passive acoustic mapping (PAM) was used to monitor cavitation emissions during treatment using a calibrated linear array. **Main Results.** It was observed that the cumulative energy of acoustic emissions during treatment was positively correlated with biofilm disruption, with differences between bacterial species attributed to differences in biofilm morphology. PCaN provided increased biofilm reduction compared to microbubbles due in large part to their persistence over the duration of ultrasound exposure. There was also good correlation between the spatial distribution of cavitation as characterized by PAM and the extent of biofilm disruption observed with microscopy. **Significance.** Collectively, the results from this work indicate the potential broad applicability of cavitation for eliminating biofilms of priority pathogens and the opportunity presented by PAM for real-time monitoring of antimicrobial processes.

1. Introduction

In addition to the ubiquitous evolution of antimicrobial resistance in bacteria, another major challenge preventing the effective treatment of bacterial infections is the presence of the biofilm: a sessile community of microbes within an extracellular polymeric substance (EPS) matrix, which imparts a number of advantages to the bacteria therein. The EPS consists of proteins, polysaccharides, and extracellular nucleic acids, which beyond providing physical protection, also enables horizontal gene transfer of resistance traits (Madsen *et al* 2012) and sophisticated methods of cell–cell communication, known as quorum sensing, that allow biofilm communities to sense and respond to environmental stressors (Rutherford and Bassler 2012). Because of this, bacteria within biofilms are reported to be between 10–1000 times more resistant to antibiotics than their planktonic counterparts (Sharma *et al* 2019) and clinically, it has been estimated that between 65% and 80% of bacterial infections involve biofilms (Joo and Otto 2012). It is demonstrable therefore that the physical and chemical properties of the EPS ensure that bacteria are largely protected from adverse environmental conditions such as UV, heat, salinity, and pH, and additionally enable shielding from antibiotics and immune cell phagocytosis (Thurlow *et al* 2011, Flemming *et al* 2016).

Due to the virulence factors associated specifically with biofilms, one therapeutic strategy that has been explored recently is targeting bacterial infections through elimination of the biofilm, such as through enzymatic degradation of EPS components (Rumbaugh and Sauer 2020), or the mechanical action of ultrasound-induced cavitation (Vyas *et al* 2019, Lattwein *et al* 2020). Cavitation, or the oscillation of a gas bubble within an acoustic field, has been hypothesized to interfere with the structural stability of biofilms, allowing either for fragmentation and dispersal of bacteria or enhanced delivery of antibiotics (LuTheryn *et al* 2020). Ultrasound cavitation either through spontaneous heterogenous nucleation (e.g. histotripsy) or introduction of external cavitation nuclei (e.g. microbubbles) have both shown efficacy at removing biofilms (Lattwein *et al* 2018, Childers *et al* 2021, LuTheryn *et al* 2022). While most work to date in this area has been performed *in vitro* or in preclinical animal models, human clinical trials of cavitation-enhanced tissue fractionation (Vidal-Jove *et al* 2022) and drug delivery (Dimcevski *et al* 2016) for oncology indicate that these effects should be translatable to biofilm applications as well. It is, however, unclear at this point whether the large acoustic intensities used in histotripsy would be required, or how the path to delivering cavitation nuclei to a site of infection may differ from the systemic or intratumoural administration used in the context of oncological drug delivery. For example, UTIs could be treated by injecting nuclei into the bladder, whilst wounds may be treated by surface or topical administration of cavitation nucleation agents. Providing guidance as to optimal future ways of treatment forms part of the objectives of the present manuscript in elucidating the underlying mechanisms.

Despite promising evidence in the literature indicating the feasibility of cavitation for elimination of bacterial biofilms, there are still several barriers to efficient clinical translation of this technique. First, previous studies testing the impact of ultrasound and microbubbles on biofilms have generally evaluated only single species of bacteria. For example, Lattwein, *et al* and Kouijzer, *et al* only tested *Staphylococcus aureus* and LuTheryn, *et al* only tested *Pseudomonas aeruginosa* (Lattwein *et al* 2018, Kouijzer *et al* 2021, LuTheryn *et al* 2022). Whether there are differences in response between species, especially between Gram-positive and Gram-negative bacteria, has yet to be established.

Second, a quantitative evaluation of the impact of cavitation ‘dose’, or the total energy of acoustic emissions measured during a therapeutic procedure (Hwang *et al* 2006, Smith and Coussios 2020), on biofilm removal has similarly not been explored. Recently, passive acoustic mapping (PAM) using calibrated ultrasound arrays has emerged as a method to provide quantitative spatial information about cavitation activity (Gray and Coussios 2018). Quantitative PAM can therefore be used to test for correlation between sensor-independent cavitation dose and biological outcome.

The key novelty of the present work is to investigate the spatiotemporal relationship between biofilm disruption and cavitation activity seeded using two different types of cavitation nucleation agents (microbubbles versus solid gas-stabilizing protein nuclei), and to characterize the acoustic emissions from cavitation activity by PAM. Quantitative monitoring of cavitation activity was compared with microscopic examination of biofilm disruption and evaluation of bactericidal effects. Two different pathogenic bacteria were included in the study: methicillin-resistant *S. aureus* (a Gram-positive bacterial species) and *P. aeruginosa* (a Gram-negative bacterial species), both of which are on the World Health Organization’s priority list for new antibiotic research and development and are known to form robust biofilms (Mancuso *et al* 2021).

2. Methods

2.1. Cavitation nuclei fabrication

A variety of cavitation agents have been explored for ultrasound-mediated therapies that offer advantages and disadvantages according to the specific application (Lyons *et al* 2023). For this work, we chose to use both microbubbles and sub-micron protein cavitation nuclei (PCaN) to promote cavitation activity. For microbubble fabrication, a 95:5 molar ratio of DPPC (1,2-dipalmitoyl-sn-glycero-3-phosphocholine) and DSPE-PEG-2k (1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000]) (Avanti Polar Lipids Inc, AL, USA) were dissolved in deionized water, propylene glycol, and glycerol (7:2:1 by volume), forming a total lipid concentration of 0.75 mg ml⁻¹. Perfluoropentane was added to the headspace and the microbubbles were formed by agitation. Microbubble size and concentration were measured with a Multisizer 4 (Beckman Coulter, Brea, CA, USA) and determined to have a median diameter of 1.2 ± 1.4 µm and concentration of 3.6 × 10⁹ microbubbles ml⁻¹, consistent with prior studies (Keller *et al* 2021). PCaN were formulated as described in (Hettinga *et al* 2023); briefly, 1 ml of chloroform containing PLA Resomer 203 H at 5 mg ml⁻¹ was overlaid with 3 ml of a 5% aqueous solution of bovine serum albumin. Probe sonication (Q-Sonica 125, Qsonica LLC, CT, USA) at 50% amplitude was then applied at the interface of the

biphasic solution for three minutes and the resulting emulsion was lyophilized to dryness, before being rehydrated in water prior to use. Particles were characterized using a Videodrop (Myriade, Paris, FR) and had a typical concentration of 6.62×10^{11} particles ml^{-1} and a median diameter of 215 nm.

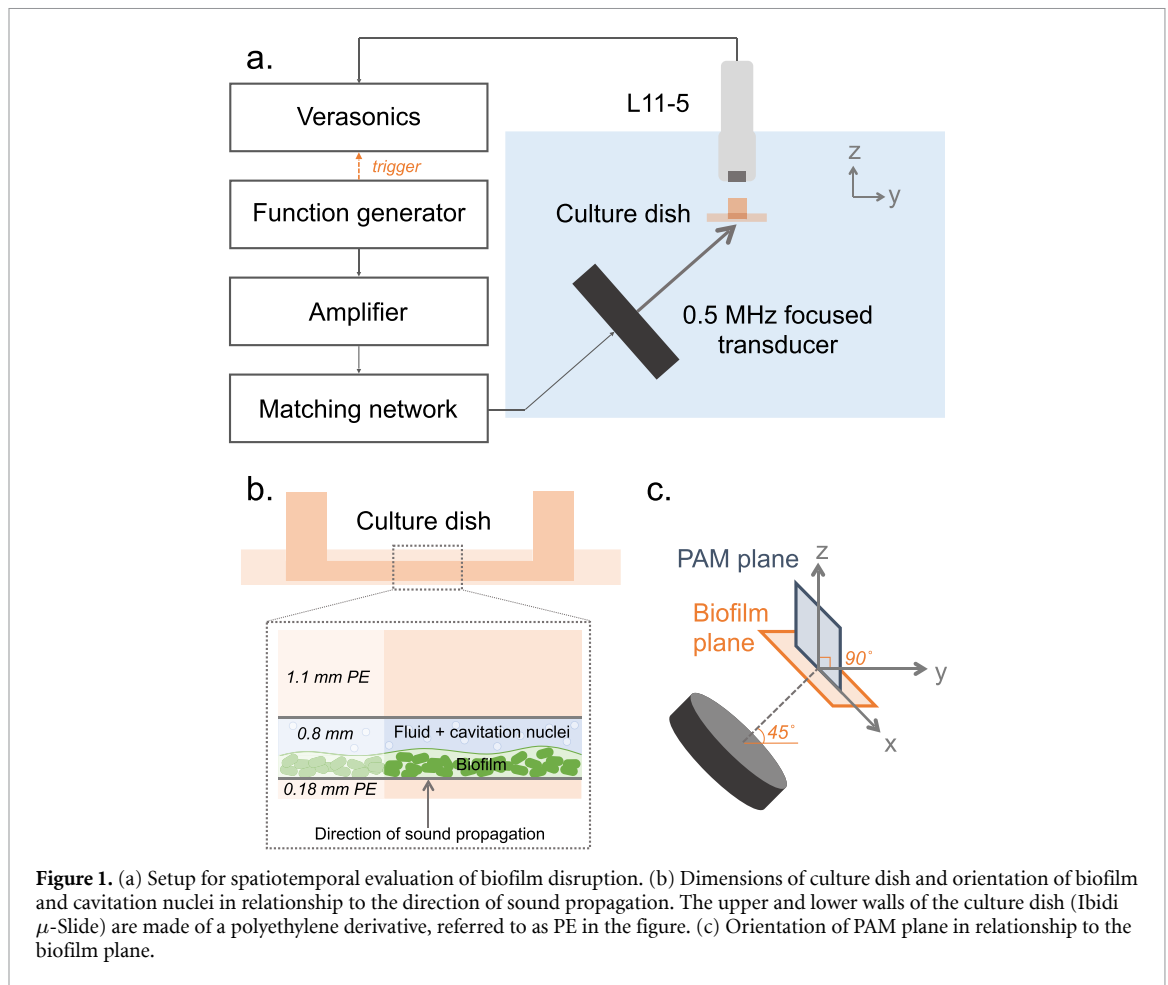
2.2. Biofilm culturing within the μ -Slide

The *S. aureus* strain was a clinical osteomyelitis isolate obtained from the Nuffield Department of Medicine at the University of Oxford. The *P. aeruginosa* (PAO1) strain was isolated from a chronic wound and is a laboratory reference strain commonly used for biofilm studies (Wilson *et al* 2017). The process for growing each strain into a biofilm was the same with exceptions noted. Bacteria were first streaked onto a Luria Bertani (LB) agar plate from frozen stock. After overnight incubation at 5% CO_2 and 37 °C, three individual colonies were selected using a sterile loop and inoculated into broth: LB broth for *P. aeruginosa*, and brain heart infusion broth for *S. aureus*. After another overnight incubation in a shaking incubator at 37 °C and 180 rpm, the bacteria were diluted 1:100 in the chosen bacterial broth and incubated at 180 rpm until the OD_{600} reached 0.2–0.3, corresponding to approximately 4×10^7 CFU ml^{-1} for *S. aureus* and 1×10^8 CFU ml^{-1} for *P. aeruginosa*. The bacteria were then added to a sterile 0.8 mm ibiTreat polymer μ -Slide (Ibidi GmbH, Gräfelfing, Germany) which had been pre-coated with 145 $\mu\text{g ml}^{-1}$ collagen I in 0.1% sterile-filtered acetic acid to aid attachment (Birkenhauer *et al* 2014) and to mimic a realistic tissue substrate as collagen I is the most abundant protein in the extracellular matrix (ECM) (Ricard-Blum 2011). The biofilms were cultured for 48 h in the μ -Slides to allow for growth into a ‘mature’ biofilm (Li *et al* 2020). *S. aureus* is a facultative anaerobe and was therefore cultured statically, whereas *P. aeruginosa* is a facultative aerobe and was thus cultured under 80 rpm agitation.

After 48 h culturing, the biofilms were gently flushed with 1 ml PBS, corresponding to approximately three times the volume of the μ -Slide, and live/dead stained with 10 μM Syto 9 and 60 μM propidium iodide according to the manufacturer’s instructions (FilmTracer Live/Dead Biofilm Viability Kit, Molecular Probes, Invitrogen). The biofilms were imaged at 10x and 40x magnification using an inverted fluorescent microscope (Eclipse Ti, Nikon Inc, USA) in 3×3 stitches, resulting in 5×5 mm and 1.25×1.25 mm large images, respectively. The 10x image was used for visualization of the spatial extent of biofilm removal while the 40x image was used for quantification of biofilm coverage area before and after ultrasound exposure. After ultrasound exposure, the supernatant containing any dispersed bacteria from the μ -Slides was collected for a viability (CFU) assay and replaced with sterile broth. The biofilms were then re-imaged on the fluorescent microscope following the protocol outlined above.

2.3. Ultrasound treatment of biofilms within μ -Slides

After imaging, cavitation nuclei (final concentration of 1×10^8 microbubbles ml^{-1} diluted into sterile water or 1×10^{10} PCaN ml^{-1} diluted into sterile water) or sterile broth alone (for US only and control groups) was added to the biofilms. The different concentrations were chosen to match the peak energy of radiated acoustic emissions (in preliminary experiments) as a proxy for matching cavitation activity. It turned out that these concentrations resulted in gas void fractions that were on the same order of magnitude (calculated based on the diameter and concentration of the particles). Three replicates were performed for each condition. The different concentrations were chosen as it was empirically determined that these concentrations provided similar orders of magnitude of maximum cavitation activity and they provided similar gas volume fractions, calculated based off particle number and radius (corresponding to the approximate bubble radius). Although high microbubble concentrations (greater than 1×10^6 microbubbles ml^{-1}) are expected to cause acoustic attenuation (Keller *et al* 2019), the orientation of the μ -Slide ensured that microbubble attenuation only affected the receive pathway and not the pressure experienced by microbubbles near the biofilm. μ -Slides were submerged in the water tank (figure 1) which was filled with degassed, deionized water at 25 °C (representative of ambient conditions for wound applications) where they were exposed to 0.5 MHz ultrasound, with 0.5–1.5 MPa peak rarefactional pressure (derated), 200 cycles, at a 5 Hz pulse repetition frequency (PRF) for 30 s generated by a spherically focused transducer (H107, Sonic Concepts, Bothell, WA, USA) with a 64 mm aperture diameter and 63 mm radius of curvature. These pulses were generated by a Keysight 33 600 A Waveform Generator (Keysight Technologies, Santa Rosa, CA, USA) and amplified by an E&I 1020 I RF Amplifier (Electronics & Innovation, Ltd, Rochester, NY, USA). To evaluate the effect of ultrasound without any added cavitation nuclei, just the highest pressure (1.5 MPa) was tested. Sham-treated controls were also included in which the biofilms were placed in the water tank for the same duration of time but without ultrasound exposure. The amplitude and pulse length were chosen so that broadband noise would be generated from both microbubbles and PCaN (Hettinga *et al* 2023) and the PRF so that there was enough time between pulses to transfer the acoustic emission data to the computer.



2.4. Cavitation post-processing for PAM

Cavitation signals were passively received by a calibrated L11-5 (Verasonics, Kirkland, WA, USA) linear array located 28 mm above the biofilm for quantitative mapping of acoustic emissions over time. All cavitation signals collected by the Verasonics system were imported into Matlab (The Mathworks, Inc., Natick, MA, USA) for further analysis. The radiofrequency (RF) voltage data were converted to pressure by means of a substitution calibration (Gray and Coussios 2018). For each PAM frame, a bespoke 50 kHz multiband finite impulse response (FIR) bandpass filter was created to separate harmonic and broadband emissions within the windowed waveforms for each channel. The filter was created by analyzing the spectrum and identifying harmonic tones as peaks that were at least 6 dB above the local background (the surrounding 100 kHz band of frequencies) (figure S1). The signals were then bandpass-filtered based on the measured sensitivity of each element, ranging from 3–6 MHz for the outer elements and 3–11 MHz for the inner elements, to avoid introducing errors into the beamforming algorithm. The calibrated waveforms were then beamformed using Robust Capon Beamforming (Coviello *et al* 2015), using an epsilon factor of 0.1, a reasonable choice for a calibrated linear array.

Emissions received during each transmit pulse were processed to yield one PAM image, in which each pixel represents an estimate of the acoustic energy emitted from that location in μ J. The processing includes correction for the pixel-to-array element distances and integration over the acquisition duration, as described in (Coviello *et al* 2015). To enable quantification of cavitation activity for each individual replicate, the sum of all pixels within each PAM frame was calculated, and is referred to in subsequent analyses as E_{total} , $E_{\text{broadband}}$, or E_{harmonic} based on whether total energy (e.g. no multiband filter), broadband emissions, or harmonic emissions were included. The sum of all E_{total} , $E_{\text{broadband}}$, or E_{harmonic} values within a complete exposure (referred to as frame integral or FI) was calculated as a measure of the total amount of cavitation experienced by the biofilm over the complete exposure (Smith and Coussios 2020) and was evaluated for correlation with biofilm reduction (BR). The frame peak (FP) or the maximum value of E_{total} , $E_{\text{broadband}}$, or E_{harmonic} over the exposure duration was also calculated as a measure of comparison between agents. It is worth noting that E_{total} is approximately equal to the sum of $E_{\text{broadband}}$ and E_{harmonic} in all cases, achieving

conservation of radiated acoustic energy to within 15% for microbubbles and 2% for PCaN accounting for minor losses due to filtering and associated energy leakage.

As a method of comparing the spatial extent of PAM energy with BR, all frames acquired within a complete exposure were summed into a single PAM image, representing the spatial accumulation of acoustic energy. For this analysis, just total (unfiltered) energy was evaluated. The summed PAM images were used to compare with the width of biofilm removal observed with microscopy.

2.5. Quantitative analysis of biofilm reduction

In order to determine the percent reduction in biofilm coverage area due to ultrasound, a process of quantitative image analysis on all 40x large image stitches was performed using a custom-made Matlab script (figure 2). Only the Syto 9 (live) channel was evaluated for coverage area reduction. After importing the images into Matlab, background subtraction was performed on the image using the rolling ball algorithm with a ball radius of 50 pixels. Then, the image was auto-thresholded using the triangle threshold method, which utilizes the histogram of intensities to find an optimal upper threshold bound (Zack *et al* 1977) to allow for maximum dynamic range of intensity values prior to binarization. If significant image saturation was observed visually even after thresholding, the upper bound was slightly increased manually. The image was then binarized, with all clusters of nonzero pixels under 20 pixels excluded as noise. This was performed on the images taken before and after ultrasound exposure. The BR was defined by the change in area:

$$BR = 1 - \left(\frac{\sum P_A}{\sum P_B} \right). \quad (1)$$

Where P_A refers to non-zero pixels in the image after ultrasound, and P_B refers to non-zero pixels in the image before ultrasound. Importantly, BR provides a measure of biofilm (and thus, bacterial) removal, not necessarily bacterial kill.

2.6. Colony-forming units assay

The vitality of the bacteria that had been removed from the surface was measured by assaying the fluid in each μ -Slide after ultrasound exposure. The fluid with any released bacteria was transferred to separate sterile Eppendorf tubes and then spun down at 4,500 g for 6 min. The supernatant was discarded, and the bacteria pellet was resuspended in 1 ml of sterile broth and agitated using a vortex mixer. Bacterial cell counts were then determined using the Miles–Misra method (Miles *et al* 1938). Briefly, the bacteria were serially diluted by 10^8 in a 96-well plate by transferring 20 μ l of each dilution into 180 μ l of sterile PBS (e.g. serial 1:10 dilutions) eight times. Three 10 μ l drops of each dilution were added to a sterile LB agar plate and placed in an incubator overnight (5% CO₂ and 37 °C). The next morning, the number of visible colonies for each dilution was counted and multiplied by the dilution factor to determine the number of colony-forming units per ml (CFU ml⁻¹).

2.7. Statistical analysis

A 2-way ANOVA was performed to find whether there were significant differences in BR both between species and within species between ultrasound conditions. Multiple comparisons unpaired *t*-tests were then performed to find differences between species. For differences within a single species between ultrasound conditions, a Dunnett's multiple comparison test was performed to determine whether there were any ultrasound parameters that resulted in BR that was significantly different to the control. In order to determine whether there was a correlation between the frame integral (FI_{total}, FI_{broadband}, or FI_{harmonic}) and biofilm removal, a Spearman's nonparametric correlation test was performed on the data for each species and each type of cavitation nuclei. For all analyses, the significance level was set to 0.05.

3. Results

3.1. Acoustic characterization of biofilm exposure conditions

The 2D acoustic field in the x - y plane measured with a fibre-optic hydrophone (Precision Acoustics Ltd, Dorchester, UK) that was threaded into the μ -Slide is shown in figure 3, with both maps normalized to the maximum pressure in the free field. The free field measurement (figure 3(a)) shows an elongated focal region, a result of the 45° angle of the scan plane relative to the transmit source. The field within the μ -Slide (figure 3(b)) shows two high-pressure regions separated by a valley at $y = 0.5$ mm. The valley is a result of sound reverberations between the upper and lower interfaces of the channel. The maximum pressure was still comparable to the free field measurements, indicating minimal attenuation through the lower surface of the μ -Slide. Notably, the μ -Slide was positioned such that the acoustic wave traveled through the biofilm first and then through the surrounding media with or without cavitation nuclei (figure 1(b)), as the upper wall of

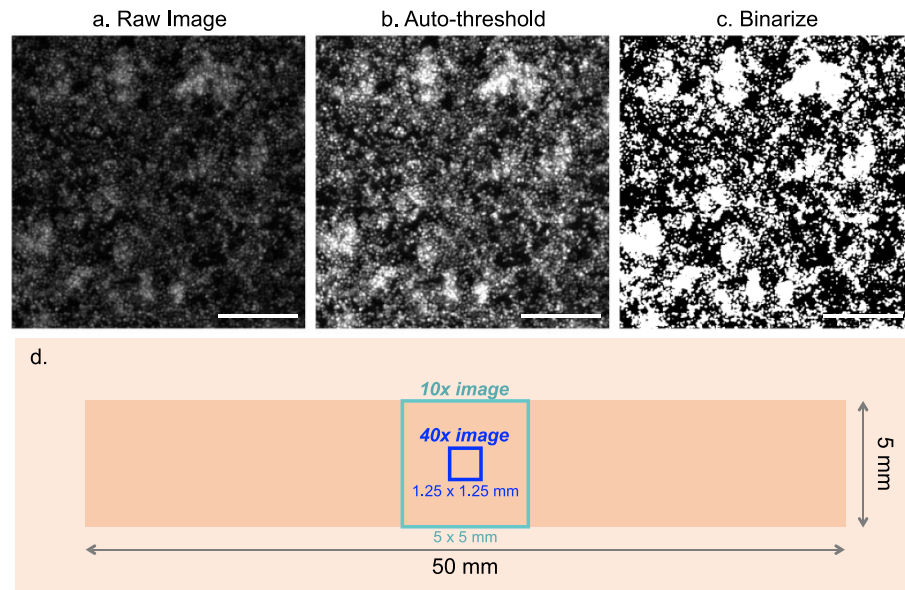


Figure 2. Zoomed-in view of the image analysis process on *S. aureus* bacteria stained with Syto 9. (a) Raw grayscale image taken from the microscope. (b) Autothresholded image (using the triangle threshold method). (c) Final binarized image from which coverage area can be derived. Scale bar represents 25 μm . (d) Size of the 10x and 40x large image stitches compared to the overall dimensions of the μ -Slide.

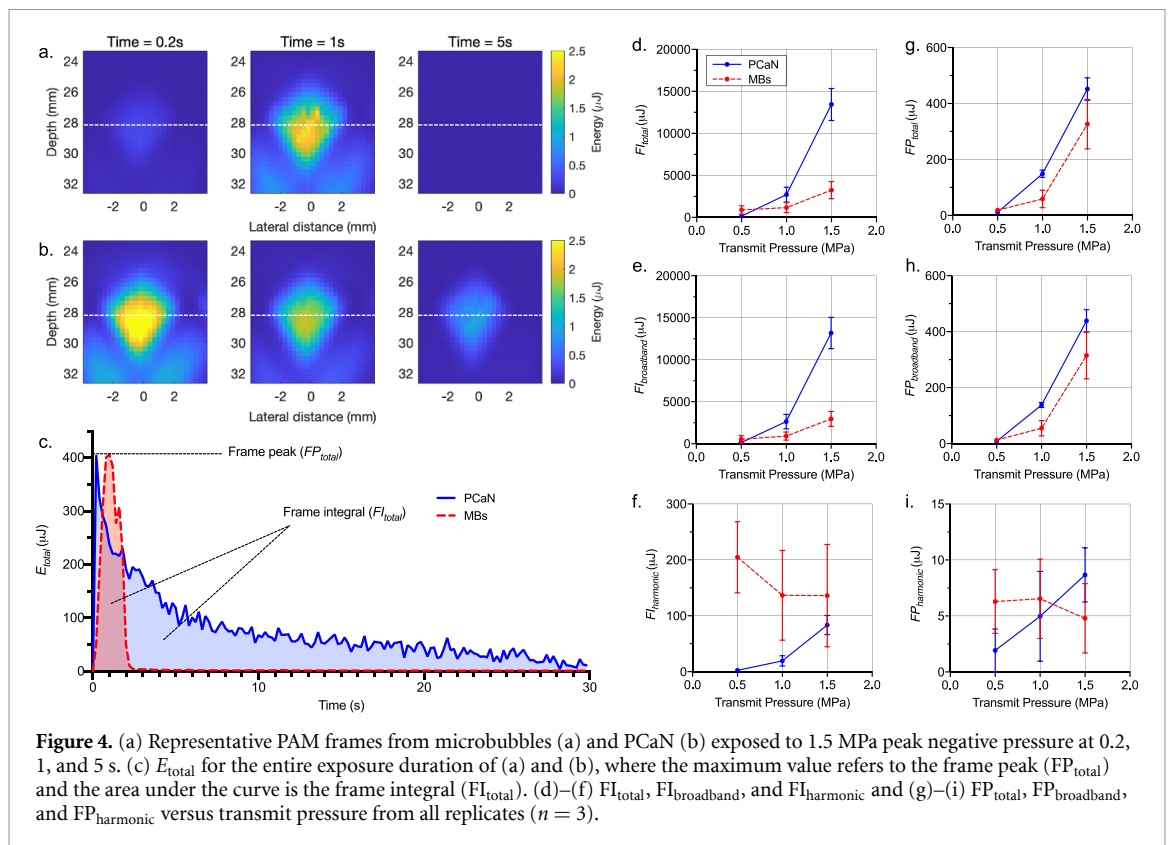
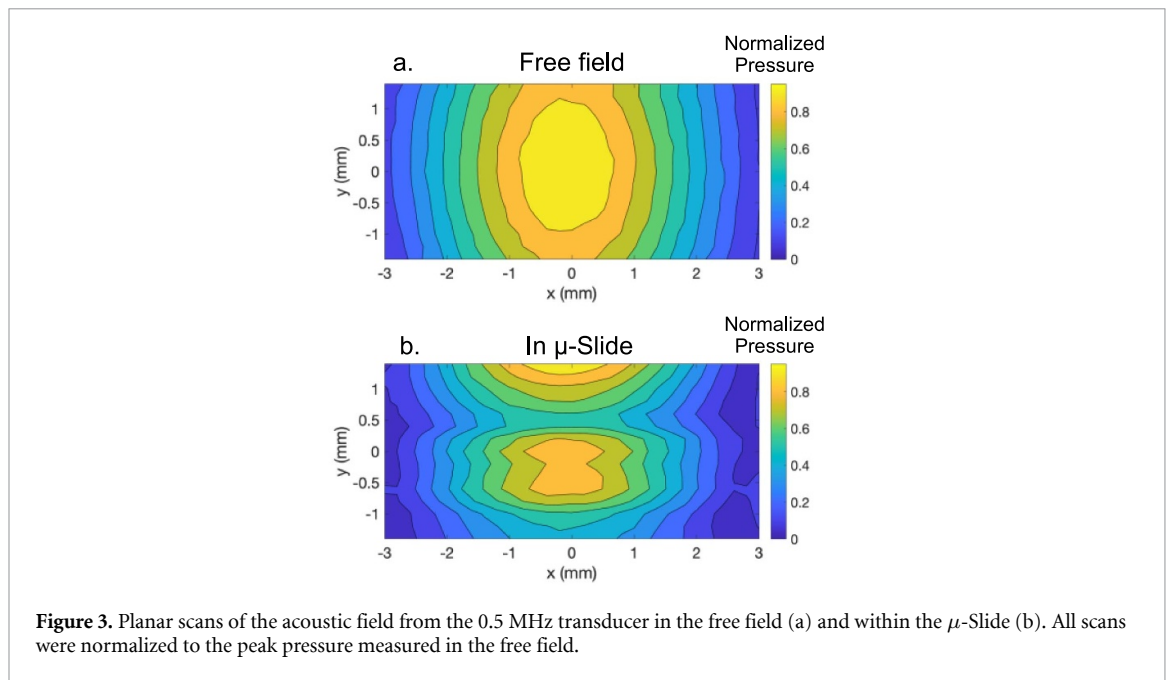
the μ -Slide, made of a polyethylene derivative, was thicker than the lower wall (1.1 mm versus 0.18 mm) and would have resulted in increased transmission loss as well as increased refraction. This is therefore a different orientation compared to that used in previous studies with μ -Slides (Kouijzer *et al* 2021, Zangabad *et al* 2023).

3.2. Effect of cavitation nuclei on intensity, persistence and extent of cavitation activity

Figure 4 shows PAM frames at 0.2, 1, and 5 s into a total 30 s exposure from representative samples taken at 1.5 MPa with either MBs (a) or PCaN (b) without multiband filtering (e.g. total energy), and (c) shows the resultant E_{total} values over the exposure duration, with FP and frame integral (FI) measurements indicated. The white dashed line in (a) shows the placement of the biofilm at 28 mm. Although the FP achieved by MBs versus PCaN is similar, the duration of cavitation activity was prolonged with PCaN and thus the FI is higher. (d)–(i) shows a comparison of the FI and FP produced by the two different cavitation nuclei at each transmit pressure for all replicates without multiband filtering or with multiband filtering where broadband and harmonic energies are separated. The unfiltered (total) energy and broadband filtered energy trends are similar; increasing transmit pressure results in increased acoustic energy, and while PCaN produced an overall greater FI both in terms of total and broadband energy, the FP from any individual transmit event was similar, which was expected given the preliminary experiments performed to choose particle concentration. From these results, it can be inferred that the inertial cavitation threshold for both agents under these conditions, demonstrated by a sharp increase in broadband acoustic emissions in (e) and (h), occurs between 1 and 1.5 MPa. It is worth noting that harmonic energy typically represented less than 2.5% of the integral of radiated acoustic energy. In that context, microbubbles excited at the lowest transmit pressure resulted in the greatest $\text{FI}_{\text{harmonic}}$, although similarly, the $\text{FP}_{\text{harmonic}}$ achieved was not significantly different between the two cavitation nuclei. For these comparisons, the total particles per ml were not the same; the concentration of microbubbles was 1×10^8 particles ml^{-1} and PCaN was 1×10^{10} particles ml^{-1} ; however, the void fraction (calculated based on the diameter and concentration) was similar.

3.3. Qualitative analysis of biofilm reduction

Figure 5 shows example microscopy images (40x) stained with Syto 9 (green, live) and PI (red, dead) from all exposure conditions. The control images (figures 5(g) and (n)), taken after sham treatment, show a uniform coverage area of bacterial cells and clusters. Increasing pressure for each condition corresponded to an overall increase in BR, although variations in the pattern of biofilm removal were observed. At 0.5 and 1 MPa, *S. aureus* biofilms remained attached to the surface with only a partial reduction in biofilm density and fewer cell clusters, but at 1.5 MPa with PCaN, most bacteria were removed. At the same pressures, *P. aeruginosa* biofilms were found to be more susceptible to detachment, shown most clearly in the differences between *S.*



aureus and *P. aeruginosa* at 1 MPa with either MBs or PCaNs (figures 5(b), (i), (e) and (l)). Furthermore, *P. aeruginosa* removal occurred mainly in localized regions, possibly a result of bubble movement along the surface of the biofilm, versus the more heterogeneous nature of bacterial removal in *S. aureus* biofilms.

3.4. Quantitative analysis of biofilm reduction

Figures 6(a)–(c) show the results of the quantitative image analysis of biofilm disruption, which provide additional evidence to support the qualitative observations shown in figure 5. An increase in peak rarefactional pressure resulted in an increase in BR. The use of PCaNs resulted in a greater degree of BR than microbubbles in both species tested at 1 and 1.5 MPa (b), (c). The conditions that resulted in BRs that were significantly different than their respective controls (for each species) were 1.5 MPa with microbubbles

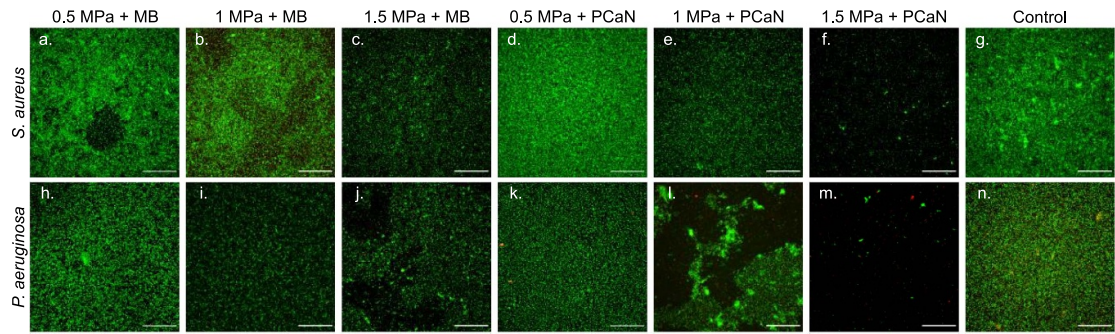


Figure 5. Representative 40x microscopy images of biofilms exposed to 0.5, 1, and 1.5 MPa peak negative pressure in the presence of either microbubbles (MBs) or protein cavitation nuclei (PCaN) in *S. aureus* and *P. aeruginosa* biofilms. The biofilms were stained with Syto 9 (green, live bacteria) and propidium iodide (red, dead bacteria). Increasing ultrasound pressure results in decreased biofilm presence, more pronounced with PCaN than with MBs. (g) and (n) show control (sham-treated) biofilm images as a comparison. Scale bars represent 100 μm .

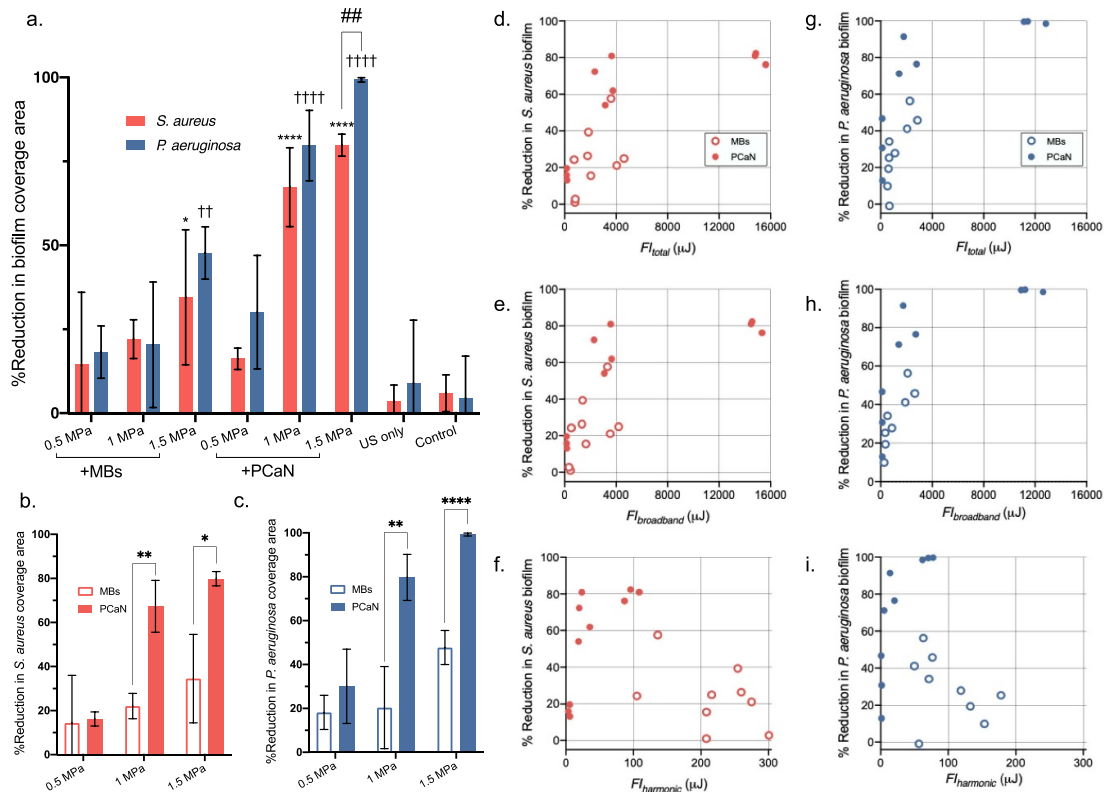
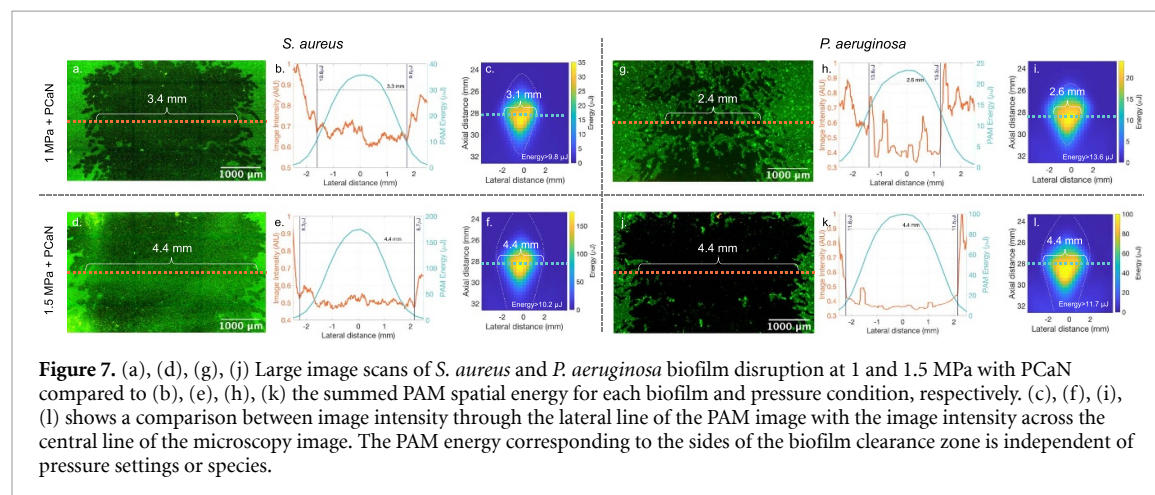


Figure 6. Quantitative evaluation of biofilm removal. (a) Percent reduction in biofilm presence for each condition evaluated and for each species ($n = 3$ replicates per condition); US only refers to 1.5 MPa without any added cavitation nuclei. * = difference between *S. aureus* experimental conditions and *S. aureus* control (** $p < 0.01$, *** $p < 0.0001$), † = difference between *P. aeruginosa* experimental conditions and *P. aeruginosa* control (†††† $p < 0.0001$), # = difference between *S. aureus* and *P. aeruginosa* conditions (## $p < 0.01$, #### $p < 0.01$). (b)–(c) show the same data as (a) rearranged to show the comparison between biofilm reduction resulting from microbubbles or PCaN for each species. * = difference between MBs and PCaN (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$). (d)–(f) and (g)–(i) Scatter plots between %biofilm reduction and Fl_{total} , $Fl_{\text{broadband}}$, and Fl_{harmonic} in *S. aureus* (red) and *P. aeruginosa* (blue) biofilms, respectively, with MBs (open marker) and PCaN (closed marker).

(34.5% $\pm$ 20.1% for *S. aureus* and 47.8% $\pm$ 10.7% for *P. aeruginosa*), 1 MPa with PCaN (67.3% $\pm$ 13.7% for *S. aureus* and 79.7% $\pm$ 14.3% for *P. aeruginosa*), and 1.5 MPa with PCaN (79.8% $\pm$ 3.2% for *S. aureus* and 99.3% $\pm$ 0.1% for *P. aeruginosa*). Although, on average, there was a greater proportion of biofilm removal observed in the *P. aeruginosa* biofilms compared to the *S. aureus* biofilms, this was only significantly different at 1.5 MPa with PCaN.

Table 1. Correlation between Cumulative FI_{total} , $FI_{broadband}$, and $FI_{harmonic}$ and reduction in biofilm surface area.

	Cavitation nuclei	Spearman's correlation coefficient r		
		FI_{total}	$FI_{broadband}$	$FI_{harmonic}$
<i>S. aureus</i>	MBs	0.38	0.48	−0.17
	PCaN	0.82**	0.81**	0.89**
<i>P. aeruginosa</i>	MBs	0.78*	0.77*	−0.37
	PCaN	0.87**	0.87**	0.93**



3.5. Effects of increasing energy of acoustic emissions on biofilm coverage area

There was a positive correlation observed between both FI_{total} and $FI_{broadband}$ and biofilm removal, significant in all cases except for *S. aureus* with microbubbles (figures 6(b), (c), (e) and (f) and table 1). In contrast, there was no correlation between $FI_{harmonic}$ and biofilm removal for microbubbles for either species (figures 6(d) and (g)). The correlation between $FI_{harmonic}$ and biofilm removal was positive for PCaN for both species, although the actual energy values were much lower and were clearly a result of bleed-through from the fundamental, second harmonic and broadband frequencies, as seen in figure S1. Collectively, these results indicate that biofilm removal is correlated with broadband acoustic emissions that are associated with violent cavitation activity, although it is relevant to note that under the exposure conditions tested, the harmonic emissions were negligibly small relative to the broadband energy.

3.6. Passive acoustic maps correlate with overall biofilm destruction area

Only the highest two pressures (1 and 1.5 MPa) with PCaN produced substantial biofilm removal, as can be seen in the microscopy images in figure 7. The corresponding cumulative PAM images are also shown in figure 7, *S. aureus* in figures 7(a)–(f) and *P. aeruginosa* in figures 7(g)–(l). For this analysis, only the total energy was evaluated. For each condition, the lateral PAM line at 28 mm (the location of the biofilm in the axial plane) was compared with a smoothed image intensity profile through the central region of the microscopy image (figures 7(b), (e), (h) and (k)). The boundaries of the biofilm removal area were determined manually and used to identify the lowest accumulated energy that still resulted in removal, represented by the two vertical grey lines, which were averaged to get a single value per replicate and then plotted against its respective destruction area, shown in figure 8. It was determined that the minimum accumulated energies that still resulted in BR for *S. aureus* were $8.9 \pm 1.8 \mu J$ and $10.7 \pm 2.1 \mu J$ and for *P. aeruginosa* were $11.0 \pm 2.4 \mu J$ and $13.6 \pm 2.3 \mu J$, for 1 MPa and 1.5 MPa, respectively ($n = 3$ for each condition). There were no significant differences observed in minimum spatially integrated energy required to remove bacteria regardless of species and ultrasound transmit condition. Within the PAM images shown in figure 7, contour lines were created at each of the PAM energy minimums calculated from the boundaries of biofilm clearance. In all experiments, the width of the PAM energy contours closely overlapped with the region in the optical images where the pixel intensity was substantially reduced, at 28 mm in the axial direction of the PAM image (corresponding to the location of the biofilm). This suggests that PAM could potentially be used as a predictor of biofilm removal.

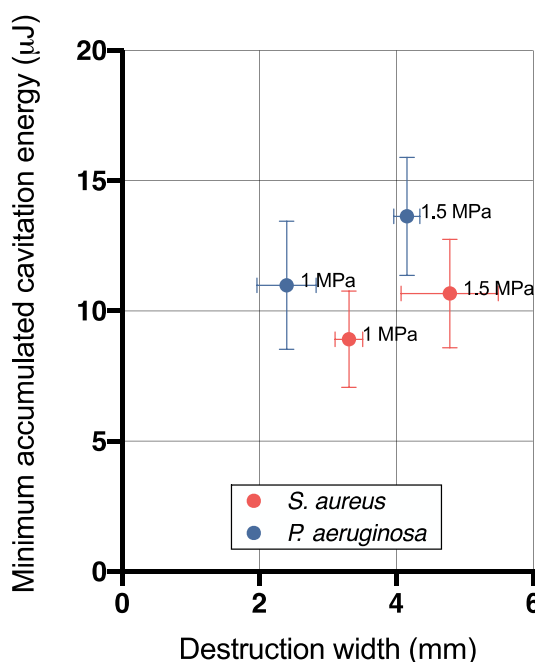


Figure 8. Comparison of the minimum accumulated cavitation energy calculated as shown in figure 7 required to remove biofilms and the resulting biofilm destruction width at the two highest pressures with PCaN ($n = 3$ replicates per condition). No significant differences were seen in cavitation energy.

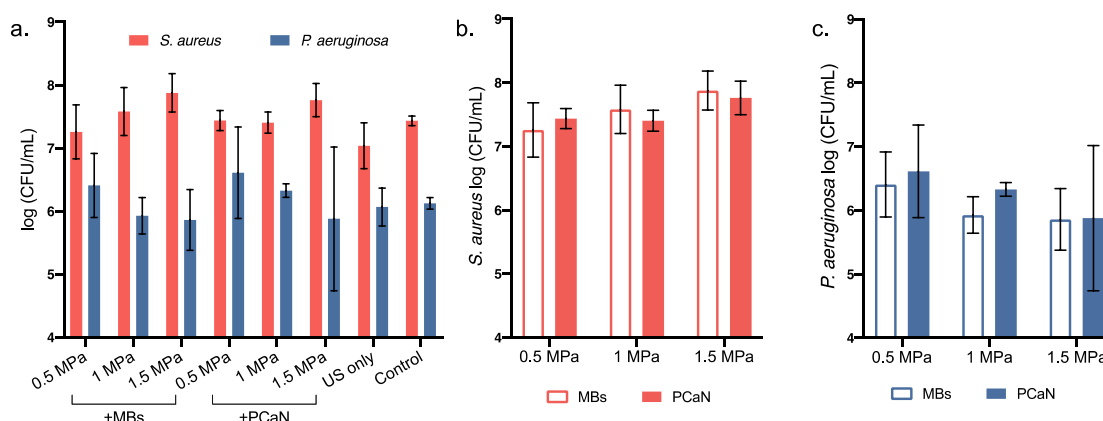


Figure 9. Colony forming units (CFU) of released bacteria in *S. aureus* and *P. aeruginosa* biofilms after exposure to all ultrasound settings (a) and rearranged to show differences between agents for each species (b), (c) ($n = 3$ replicates per condition). No significant differences were observed between each condition and its respective control. US only refers to 1.5 MPa without any added cavitation nuclei.

3.7. Colony-forming units assay

After ultrasound exposure, released bacteria from the μ -Slide were recultured to evaluate their viability. It was observed that there were no significant changes in CFU ml^{-1} in any treatment condition relative to the control for each respective species (figure 9). The CFU ml^{-1} of released bacteria from *S. aureus* biofilms was in general higher than *P. aeruginosa* biofilms (including for the controls) indicating a higher overall bacterial load within the biofilm.

4. Discussion

The goal of this work was to investigate ultrasound-induced cavitation to disrupt bacterial biofilms as a form of antibiotic therapy. PAM was used as a method of treatment monitoring. We observed a correlation between both total and broadband cumulative cavitation emissions and biofilm removal for all conditions except *S. aureus* exposed to microbubbles. In addition to quantitative energy data, PAM was also able to accurately resolve the spatial area of biofilm removal which could be used to guide real-time treatment

monitoring. These results are promising for the applicability of this technique to treat biofilms of clinically significant pathogens.

Both microbubbles and PCaN produced comparable maximum cavitation energies, but PCaN showed greater persistence of cavitation over repeated insonations over the 30 s treatment exposure compared with microbubbles under the acoustic conditions selected for these experiments. This is potentially because, by virtue of their smaller size, the concentrations of PCaN can be several orders of magnitude higher than microbubbles, or possibly because as solid cavitation nuclei they are less prone to destruction. Collectively, this work indicates that the main benefit of PCaN for this application is their persistence and that their size permits more cavitation nuclei per unit volume, which therefore increases the probability of a cavitation event happening next to an individual bacterium, e.g. the so-called ‘bubble-to-cell’ ratio used elsewhere (Guzmán *et al* 2003). Microbubbles could provide a simpler path towards clinical implementation as they are already approved as ultrasound contrast agents and are in clinical trials for enhancing ultrasound therapies (Bouakaz and Michel Escoffre 2024); it is possible that microbubble infusion could achieve similar amounts of biofilm clearance, but this may be at the expense of slightly longer treatment times to allow for microbubble replenishment. Regardless of agent, the use of cavitation nuclei for antibiofilm treatment is limited to infection sites that can be easily accessed, such as chronic wounds via topical application, urinary tract infections via catheter administration, or vascular stents via intravenous injection.

Differences were observed in the cavitation energy versus biofilm removal between the two biofilm species tested. The highest FI tested resulted in $99.3\% \pm 0.7\%$ clearance for *P. aeruginosa* biofilms compared to $79.8\% \pm 3.2\%$ clearance for *S. aureus* biofilms, where removal seemed to plateau. These differences in biofilm removal may be explained by the differences in host-pathogen interactions and the overall properties of the biofilms formed. Both Gram-positive (i.e. *S. aureus*) and Gram-negative (i.e. *P. aeruginosa*) pathogens are known to create robust biofilms that adhere to ECM proteins such as collagen and fibronectin (Singh *et al* 2012, Vaca *et al* 2019, Arora *et al* 2021). Indeed, a collagen I binding protein has been well-characterized for *S. aureus* (Madani *et al* 2017), whereas while *P. aeruginosa* has been shown empirically to adhere to collagen I *in vitro* (Plotkowski *et al* 1991, Birkenhauer *et al* 2014), it also relies on other receptors, including to laminin and polysaccharides, to aid attachment (Singh *et al* 2012, Paulsson *et al* 2019). It is possible therefore that the adherence of *S. aureus* was stronger to the collagen I precoating. Similarly, different bacterial species (and indeed different strains of a single species) have been known to produce varying amounts of EPS, which may also impact the adherence of the bacteria both to the substrate and to the rest of the biofilm (Tallawi *et al* 2017). Biofilms with more EPS, likely PAO1 in this study, might then prove more susceptible to cavitation as they can be detached in bulk rather than in a cell-by-cell manner as was perhaps the case for *S. aureus*, although as a bactericidal strategy, a cell-by-cell method might be more effective. This may also help to explain why there was not a significant correlation between cavitation dose and biofilm removal observed when using microbubbles on *S. aureus* biofilms, as the total number of cavitation nuclei (e.g. bubble-to-cell ratio) and therefore probabilistic removal of individual cells was lower for microbubbles. Moreover, the morphology of the biofilms formed *in vitro* by *P. aeruginosa* and *S. aureus* may lend themselves to differing treatment outcomes. Specifically, biofilm thickness and growth uniformity can substantially alter the competence of the biofilm structure, thus altering their susceptibility to physical and chemical intervention (Suarez *et al* 2019, Fei *et al* 2020). In this work, it is plausible that the greater depth and surface irregularity of *P. aeruginosa* biofilms, compared to the more uniform, confluent growth of *S. aureus* biofilms, provides greater opportunity for cavitation nuclei to infiltrate the biofilm and elicit removal in response to ultrasound. While ultrasound is effective at clearing most of the biofilm regardless of bacterial species at certain ultrasound conditions, the specific interactions between pathogen and host may result in differences in efficacy and therefore differences in the required cavitation dose to fully treat the infection. The percentage of remaining bacteria could also inform whether an adjuvant, such as an antibiotic (Kouijzer *et al* 2021) or bioactive gas (LuTheryn *et al* 2022), might be a necessary part of a complete treatment as they are likely to be more effective against single cells rather than biofilms.

In addition to quantitative energy data, the usefulness of PAM over other methods of passive cavitation detection can be mainly attributed to its ability to resolve spatial information of cavitation (Gyongy and Coussios 2010). In this study, it was shown that the spatial extent of PAM energy was an indicator of the extent of biofilm removal. The minimum amount of spatially integrated PAM energy that coaligned with the outer edge of biofilm removal was not significantly different even at different transmit pressures, suggesting that calibrated PAM energy level might be used to predict biofilm removal. The combination of quantitative cavitation data with spatial information also allows for exciting therapeutic avenues, including using PAM as a feedback control system for treating areas larger than the focal size of the transducer.

One concern about using ultrasound to disrupt biofilms is that it could potentially result in dispersal of bacteria to form downstream infections. In this study, the CFU ml⁻¹ of released bacteria after ultrasound exposure was not shown to be significantly different than the control which suggests the released bacteria did

not infect the fluid beyond background levels. The caveat to this is that the removed area comprised between 5%–10% of the total biofilm area. Because of this, it is possible that the assay was not sensitive enough to capture any significant increases in bacteria release. Future work will involve a PAM-informed raster scan technique that could cover the entire treatment area, which should allow more conclusive evidence to support the safety of this therapeutic strategy. Additionally, it should be noted that planktonic bacteria are known to be more susceptible to antibiotics and immune cell phagocytosis than bacteria residing in biofilms (Costerton *et al* 1999, Cerca *et al* 2006). Therefore, even if cavitation does result in dispersal of viable bacteria, the released bacteria should be easier to eliminate innately by the body or with co-administered antibiotics. Additionally, cavitation could just be limited to applications such as chronic wounds or urinary tract infections in which the released bacteria could be easily washed away without the risk of bloodstream infections.

There are several opportunities to further extend and improve on this study in future work. First, all biofilms in this work were grown as monospecies cultures, whereas most clinical biofilms are diverse polymicrobial communities (Hall-Stoodley *et al* 2004). Indeed, it has been shown that multispecies biofilms can be thicker and more resistant to treatment than monospecies biofilms (Donlan 2002, Yadav *et al* 2017). However, it can be challenging to develop consistent multispecies models *in vitro* due to competition between pathogens, despite clinical evidence supporting symbiotic relationships between microbes *in vivo*, including *P. aeruginosa* and *S. aureus* (Hotterbeekx *et al* 2017, Yadav *et al* 2017). On the other hand, even in polymicrobial biofilms, specific species of bacteria may be constrained to different regional areas. For example, in infections containing both pathogens studied, *P. aeruginosa* is often found in deeper tissue whereas *S. aureus* is normally found on the infection surface (Fazli *et al* 2009). Given that focused ultrasound can be localized to very small regions, studying the susceptibility of single bacterial species to cavitation may therefore still be clinically relevant. Likewise, although only two bacterial species were compared within this work, they represent two of the most critical biofilm-related pathogens that cover both Gram-positive and Gram-negative infections. Our results indicate that a far larger body of work will be needed to generalize this across the variety of bacterial strains and associated responses that may be encountered. Finally, the ultrasound parameters evaluated were restricted to amplitude and cavitation nuclei, and the sample size was limited to $n = 3$. That said, this parameter set was sensitive enough to observe large differences both between species and between parameters, but future work could involve a broader parameter regime with a larger sample size in a more complex biofilm model to elucidate subtle differences between conditions.

5. Conclusion

The goal of this work was to investigate the use of cavitation to eliminate biofilms of two priority pathogens, *P. aeruginosa* (Gram negative) and *S. aureus* (Gram positive). PAM was utilised to provide quantitative analysis of cavitation energy. We compared two different cavitation agents, gas-filled microbubbles and solid PCaN. We found a positive correlation between cumulative cavitation energy and biofilm reduction in all cases except for using microbubbles on *S. aureus* biofilms, which we attribute to variations in biofilm morphology. The sub-micrometre PCaN produced more sustained cavitation leading to greater biofilm disruption than microbubbles, whilst also providing a strong and potentially predictive correlation between the broadband component of the PAM signal that could be used for treatment monitoring. It was determined that the minimum PAM energy required to eliminate biofilms was consistent across transmit pressures and bacterial species, suggesting real-time feedback could be employed to guide treatment. These data suggest that using PCaN for cavitation-enhanced biofilm removal is a promising form of antimicrobial therapy.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

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Conflict of interest

BL, MG and CCC are co-inventors on patent application families WO2022162395A1, WO2022162397A1, WO2022162396A1 and WO2022162399A1 on protein cavitation nuclei, which are not commercially licensed and are presently assigned to Oxford University Innovation.

ORCID iDs

Sara B Keller  <https://orcid.org/0000-0001-5087-4775>
 Gareth LuTheryn  <https://orcid.org/0000-0002-5383-1081>
 Michael D Gray  <https://orcid.org/0000-0002-3245-3296>
 Brian Lyons  <https://orcid.org/0000-0002-2779-8732>
 Robin O Cleveland  <https://orcid.org/0000-0002-1797-4704>
 Eleanor Stride  <https://orcid.org/0000-0003-3371-5929>
 Constantin C Coussios  <https://orcid.org/0000-0001-8792-7818>

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